GANPAT UNIVERSITY

MCA SEMESTER -II EXAMINATION

P12A1FMA: Fundamental of Mobile Applications

April - June 2023

Time: 3 Hours] [Total Marks 60

Instructions:

1. **Figure to the right indicate full Marks**
2. **Each section should be written in a separate answer book**
3. **Be precise and to the point in your answer**

SECTION-I

Que 1.		**Answer the following**	**(8)**
	1.	What do You mean by intent?	1
	2.	List out all gradient's types.	1
	3.	Give names of Tools available in SDK.	1
	4.	What is Layout? Explain any two layouts in details	5
Que 2.		**Answer the following (Any Two)**	**(11)**
	1.	What is Android? Explain Lifecycle Activity with Diagram.	5 ½
	2.	What is AVD? How to configure AVD? Explain.	5½
	3.	Short note on: Architecture of Android OS	5 ½
Que 3.		**Answer the following (Any Two)**	**(11)**
	1.	Elaborate: Components of Android OS with its skeleton code.	5 ½
	2.	Explain about the following basic view A) Image Button B) Check Box C) Toggle Button D) Radio Group E) Radio Button,.	5 ½
	3.	List out any four types of layouts built-in to the Android SDK.	5 ½

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GANPAT UNIVERSITY
M.C.A. SEM-II EXAMINATION (CBCS)
P12A2WTD WEB TECHNOLOGY USING .NET
April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following. **(8)**

(i) Define the term: Class and Object

(ii) Describe the term Exception Handling.

(iii) Define the term: Constructor and Destructor

(iv) What is an inheritance? Write down a syntax inheritance.

2. Answer the following: (Any Two) **(11)**

(i) Explain the jagged array with a suitable example.

(ii) Write a short note on One Dimensional Array.

(iii) What is conditional Statement? Explain the Switch Statement with an example.

3. Answer the following: (Any Two) **(11)**

(i) Write a short note on Method Overloading.

(ii) What is Collection? Explain HashTable with example.

(iii) Explain any two Query Operators with example.

SECTION-II

4. Answer the following. **(8)**

(i) Describe the advantage and disadvantages of ASP.NET Core MVC.

(ii) What is View Engine? List out the type of View Engine.

(iii) What is Routing?

(iv) Describe the term : Temp Data.

5. Answer the following: (Any Two) **(11)**

(i) What is View? Explain Layouts View in ASP.NET Core MVC.

(ii) What is validation? Explain the various type of validation in asp.net core mvc.

(iii) What is an Action Result? Explain various types of action results and return types.

6. Answer the following: (Any Two) **(11)**

(i) Write a short note on Razor View Engines.

(ii) What is URL Route? Explain the types of routing systems.

(iii) Write down a step to create an ASP.NET Core MVC Application.

END OF PAPER

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GANPAT UNIVERSITY

M.C.A. Semester-II Examination (CBCS)

P12A3SE : Software Engineering

April - June 2023

Total Marks: 60

Time: 3 Hours

Instructions:

1. Figures to the right indicate full marks.
2. Each section should be written in a separate answer book.
3. Be precise and to the point in your answer.

	SECTION – I	
1.	(a) Briefly describe the software engineering myth	4
	(b) Discuss the applications of software.	4
2.	**Answer the following (Any two)** 1. Discuss the generic process framework for the software engineering. 2. Explain the essence of software engineering process. 3. Write a short note on spiral model with example and diagram.	11
3.	**Answer the following (Any two)** 1. Discuss the generic concept and principles that apply to the framework activity. 2. Explain various categories of risk with proper justification. 3. Write a short note on (Risk Mitigation, Monitoring and Management) RMMM.	11

	SECTION – II	
4.	(a) what do you mean by UML(Unified Modeling Language)? Explain where UML can be used.	4
	(b) Explain the Common Mechanisms in the UML.	4
5.	**Answer the following (Any two)** 1. Discuss with suitable example using UML diagram for (i) Association (ii) Dependency (iii) Aggregation and (iv) Generalization 2. Explain interface, type and role with suitable example. 3. Briefly describe Use Case Diagram. Draw Use Case Diagram of Online student management system.	11
6.	**Answer the following (Any two)** 1. Draw a Class Diagram, Sequence Diagram and Activity Diagram of Online Purchase System 2. Discuss collaboration diagram with a suitable example. 3. Explain Swimlane diagram with a suitable example.	11

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GANPAT UNIVERSITY

M.C.A. SEMESTER-II EXAMINATION

April - June 2023

P12A4FST: Fundamental of Software Testing

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following. **(8)**

(1) Differentiate between Functional and Non-Functional testing.

(2) Explain Integration testing and System testing.

2. Answer the following. (ANY TWO) **(11)**

(1) Discuss the significance of software testing for improving the quality of the software.

(2) Considering an example, differentiate between verification and validation.

(3) Explain the V-Model of software testing using appropriate diagram.

3. Answer the following. (ANY TWO) **(11)**

(1) List the various software testing techniques. Explain any 2 of them with appropriate examples.

(2) Explain static White Box testing.

(3) Discuss dynamic Black Box testing.

SECTION-II

4. Answer the following. **(8)**

(1) Explain the different types of bugs.

(2) What are the different ways in which an external tool can access a bug tracking tool? Which of them is used by Mantis? Explain in brief.

5. Answer the following. (ANY TWO) **(11)**

(1) Discuss how can Usability testing help in improving the quality of a web application.

(2) What do you mean by Stress testing and Load testing? Explain its significance.

(3) "Performance testing is a testing process to determine the speed or effectiveness of software." Justify the statement.

6. Answer the following. (ANY TWO) **(11)**

(1) List the components of Bugzilla and explain their usages.

(2) Discuss the simple search and Advance Search options of Bugzilla.

(3) List the steps for reporting an issue in Mantis Bug Tracker.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.C.A. SEMESTER-II EXAMINATION

April - June 2023

P12A4CS1: CYBER SECURITY-I

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following **(8)**

(A) What is Cyber Security? Describe CIA Triads.

(B) Write a suitable example of Rail fence encryption.

2. Answer the following (Any Two) **(11)**

(A) List out Web based Attack and explain any two Web based Attack.

(B) What is virus? Describe any three System based attack.

(C) What is Cryptography? Describe any two technique of Modern (Encryption) Cryptography.

3. Answer the following (Any Two) **(11)**

(A) Describe any two substitution techniques of classical cryptography.

(B) Write a short note on: Public key infrastructure (PKI)

(C) Explain Social Engineering.

SECTION-II

4. Answer the following **(8)**

(A) Write a short note on Content Security Policies (CSP).

(B) Define : Nmap , Metasploit Framework.

5. Answer the following (Any Two) **(11)**

(A) How Input validation testing works for Reflected and Stored cross site Scripting.

(B) Explain SQL injection input validation testing.

(C) Describe counter measures for web application attacks.

6. Answer the following (Any Two) **(11)**

(A) What is Vulnerability Assessment? How vulnerability assessment is differ from Pentesting?

(B) Explain various stages of Penetration Testing.

(C) Usage of TCP three-way handshake protocol.

END OF PAPER

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GANPAT UNIVERSITY
M.C.A. SEM-II EXAMINATION (CBCS)
P12A4BD1 BIG DATA ANALYTICS-I
April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following. **(8)**

(i) Define the term : Datasets

(ii) What is Semantic analysis?

(iii) Explain the Business Intelligence in brief.

(iv) Describe the Predictive Analytics.

2. Answer the following: (Any Two) **(11)**

(i) What is Big Data? List out the advantage of Big Data

(ii) Write a short note on Key Performance Indicators.

(iii) Explain the Big Data Architecture.

3. Answer the following: (Any Two) **(11)**

(i) Explain the various type of characteristics of Big Data.

(ii) Explain the Quantitative analysis Techniques in detail.

(iii) Write a short note on Different Types of Digital Data.

SECTION-II

4. Answer the following. **(8)**

(i) Define the term : Name node and Data node

(ii) What is Data Serialization?

(iii) What is Map-reduce?

(iv) Describe the Resource Manager

5. Answer the following: (Any Two) **(11)**

(i) Explain the Hadoop Ecosystem's Core components with figure.

(ii) Write a short note on the types of Hadoop Vendors.

(iii) What is Apache Hadoop? Explain the advantages of Hadoop

6. Answer the following: (Any Two) **(11)**

(i) Write a short note on HDFS Architecture.

(ii) Explain the Map Reduce Distributed Processing.

(iii) Describe the Hadoop Yarn in Detail.

END OF PAPER

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GANPAT UNIVERSITY

M.C.A. Semester-II Examination (CBCS)

P12A5AIR Artificial Intelligence and Robotics

April - June 2023

Total Marks: 60

Time: 3 Hours

Instructions:

1. Figures to the right indicate full marks.
2. Each section should be written in a separate answer book.
3. Be precise and to the point in your answer.

SECTION – I

1. (a) What is Artificial Intelligence (AI) ? Why AI is important? 3
   (b) Write a short note on development of AI Languages. 5

2. **Answer the following (Any two)** 11
   1. Explain intelligent Agent with examples and property of environments.
   2. Discuss Sub area of Artificial Intelligence (AI).
   3. What is prolog list? Elaborate any 3 list manipulation operations with examples.

3. **Answer the following (Any two)** 11
   1. Explain Components of AI Program.
   2. Discuss Advantages and disadvantage of Prolog programming.
   3. List out and discuss Applications of AI.

SECTION – II

4. Define Actuators and discuss types of Actuators. 8

5. **Answer the following (Any two)** 11
   1. Discuss Depth-First Search with a suitable.
   2. Explain A* algorithm in detail with a suitable example.
   3. Explain components of Robotics.

6. **Answer the following (Any two)** 11
   1. Explain uninformed search strategies.
   2. Discuss applications of Robotics.
   3. Write a short note on Continuous Path Control of Robot.

----- XX ---End of the Paper ---XX -----

Enrollment No._____

GANPAT UNIVERSITY

M.C.A. SEMESTER-II EXAMINATION

P12A5STA Software Testing Automation

Time: 3 Hours] April - June 2023 **[Total Marks: 60**

Instructions to the examinee:

1. Figures to the right indicate full marks.
2. Each section should be written in a separate answer book.
3. Be precise and to the point in your answer.

SECTION-I

1. Answer the following **(8)**

(A) What is Automation Testing?

(B) What is Selenium IDE? Describe Selenium IDE Features.

2. Answer the following (Any Two) **(11)**

(A) What are the types of manual testing & how manual testing is performed?

(B) Enlist difference between manual and automation testing.

(C) Write short note on (any 2) automation testing tools:

a) Katalon Studio b) Selenium c) Appium

3. Answer the following (Any Two) **(11)**

(A) Write down steps to configure selenium Webdriver (Java or Python).

(B) Discuss Architecture of Selenium Web Driver with a neat diagram.

(C) Write short note on Selenium RC. How it is different from Selenium Webdriver.

SECTION-II

4. Answer the following **(8)**

(A) What are various navigate() commands in Selenium Java Webdriver? Explain.

(B) Write down step to configure Selenium web driver to PyCharm IDE.

5. Answer the following (Any Two) **(11)**

(A) Write down the steps to handle dynamic table in Selenium Java Webdriver?

(B) Explain steps to read Configurations from Property File in Selenium Java ?

(C) Write steps to generate reports using Selenium Java Webdriver?

6. Answer the following (Any Two) **(11)**

(A) How to Locating Elements by ID, Name, XPath and class name using Selenium Python?

(B) Write a short note on Implicit and Explicit wait for Selenium Python?

(C) What are Page Object Classes in Selenium Python? What are advantages of Page Object Model?

END OF PAPER